Diabetes Risk Prediction Patient Segmentation

Internship Report

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to

Submitted to the University of Kerala in partial fulfillment of the requirements for the FYUGP Summer Internship

BACHELOR OF COMPUTER APPLICATIONS (BCA)

Under the guidance of
....

Bachelor of Computer Applications

Sree Narayana College of Technology, Vadakkevila, Kollam

JUNE 2026

Sree Narayana College of Technology, Vadakkevila
Dept.of Computer Application

C E R T I F I C A T E

This is to certify that this report titled **................................** is a bonafide record of the **...............** presented by **................**, under our guidance and supervision, in partial fulfillment of the requirements for the award of the degree **Bachelor of Computer Applications** under **University of Kerala , thiruvananthapuram**.

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ACKNOWLEDGEMENT

I would like to express my sincere gratitude to all those who have supported and guided me throughout this internship project.

First and foremost, I am deeply thankful to **Dr. Chinnu Jacob**, HOD & Assistant Professor, Centre for AI, TKMCE, for her invaluable guidance, consistent encouragement, and expert supervision throughout this work. Her insights greatly helped in shaping the direction and quality of this report.

I extend my heartfelt thanks to **Prof. Sree S Bhagya**, Assistant Professor, Centre for AI, TKMCE, for her continuous support as Project Coordinator and for keeping the project on the right track.

I am grateful to **Dr. Salini S Nair**, HOD & Assistant Professor, Department of Computer Applications, Sree Narayana College of Technology, Vadakkevila, for providing the necessary infrastructure and academic environment to carry out this work.

I would also like to thank the faculty and staff of the Department of Computer Applications, SNCT, for their encouragement and for making the internship a productive learning experience.

Finally, I express my deepest appreciation to my family and friends for their unwavering support and motivation throughout this journey.

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ABSTRACT

Diabetes mellitus is one of the fastest-growing chronic diseases globally, affecting millions of individuals and placing a significant burden on healthcare systems. Early and accurate prediction of diabetes risk, combined with intelligent patient segmentation, can greatly improve clinical decision-making and enable personalized healthcare interventions.

This internship report presents a machine learning-based framework for diabetes risk prediction and patient segmentation using the Pima Indians Diabetes Database. The study applies a structured data preprocessing pipeline that addresses missing values through skewness-aware imputation (mean or median), followed by log transformation and outlier detection using the Local Outlier Factor (LOF) algorithm, which reduced the effective dataset from 768 to 387 high-quality instances.

Four supervised classification algorithms — Logistic Regression, Decision Tree, Random Forest, and Support Vector Machine (SVM) — were trained, evaluated, and compared using cross-validation accuracy. In addition, an unsupervised K-Means clustering approach was applied to segment patients into distinct risk groups based on their health profiles, enabling tailored treatment strategies.

Experimental results demonstrate that the Random Forest classifier achieves the highest cross-validation accuracy of **97.22%**, followed by SVM (94.44%), Decision Tree (93.52%), and Logistic Regression (87.96%). The clustering analysis further reveals meaningful patient subgroups with different risk characteristics.

The proposed system contributes to the field of intelligent healthcare by offering an integrated prediction and segmentation solution that can assist clinicians in early diagnosis and resource prioritization.

Keywords: Diabetes Prediction, Machine Learning, Patient Segmentation, K-Means Clustering, Random Forest, Local Outlier Factor, Log Transformation, Pima Indians Diabetes Database.

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Chapter 1

Introduction

1.1 Background

Table 1.1: Global Diabetes Statistics Overview

Year	Number of Cases (Million)	Projected Increase (%)
2017	424	–
2021	537	26.6%
2030 (Projected)	643	19.7%
2045 (Projected)	783	21.7%

The rapid increase in diabetes cases highlights the need for early detection and efficient predictive systems using modern technologies such as machine learning and data analytics.

Diabetes mellitus is a chronic metabolic disorder that affects millions of people worldwide and poses significant health and economic challenges. It occurs when the body is unable to properly regulate blood glucose levels due to insufficient insulin production or ineffective insulin utilization. Early detection and prediction of diabetes are essential for timely intervention, effective disease management, and reduction of long-term complications such as cardiovascular diseases, kidney failure, and neuropathy. In recent years, the rapid growth of healthcare data and advancements in machine learning and deep learning techniques have created new opportunities for improving diabetes diagnosis and prediction. These intelligent models are capable of identifying complex patterns and relationships within medical data that are often difficult to detect using traditional statistical methods [1, 2].

1.2 Problem Definition

Despite advancements in healthcare technologies, accurate and early prediction of diabetes remains a significant challenge. Many existing diagnostic approaches rely heavily on manual analysis and clinical expertise, which may lead to delayed detection and inconsistent results. Additionally, medical datasets often contain missing values, imbalanced classes, and complex feature interactions, making prediction tasks more difficult. Traditional statistical methods are limited in capturing nonlinear relationships among attributes such as glucose levels, BMI, insulin, and age. These limitations highlight the need for efficient, automated, and intelligent predictive systems that can improve diagnostic accuracy and assist healthcare professionals in decision-making [3].

1.3 Scope and Significance of the Work

The scope of this work focuses on developing a data-driven approach for diabetes risk prediction and patient segmentation using machine learning and deep learning techniques. The availability of benchmark datasets such as the Pima Indians Diabetes Database provides a strong foundation for analyzing patient health data and building predictive models. This study also emphasizes the importance of patient segmentation, which enables grouping individuals based on risk levels and health conditions, thereby supporting personalized healthcare strategies.

The significance of this work lies in its potential to improve early diagnosis and preventive healthcare. By leveraging modern data analytics techniques, the proposed system can assist healthcare providers in identifying high-risk individuals and optimizing treatment plans. Furthermore, the integration of prediction and segmentation enhances decision-making capabilities and contributes to the development of intelligent healthcare systems [4].

1.4 Objective

The primary objectives of this dissertation are:

- To develop an effective diabetes risk prediction model using machine learning
- To preprocess and analyze medical datasets for improved prediction accuracy
- To compare the performance of different classification algorithms
- To perform patient segmentation based on risk levels using clustering techniques
- To support early detection and personalized treatment strategies for diabetes patients

Chapter 2

Literature Survey

2.1 Existing Studies

Recent research in diabetes prediction has widely utilized machine learning and deep learning techniques on datasets such as the Pima Indians Diabetes Database. Zehra et al. highlighted the importance of data preprocessing and showed that proper preprocessing significantly improves classification accuracy [4]. Chang et al. developed an e-diagnosis system using interpretable machine learning models such as Naive Bayes, Decision Tree, and Random Forest, emphasizing transparency in healthcare decision-making [1].

Deep learning approaches have also shown strong performance in diabetes prediction tasks. Mousa et al. compared LSTM, CNN, and Random Forest models and reported that LSTM achieved the highest accuracy due to its ability to capture complex data patterns [2]. Similarly, Reza et al. proposed an improved Support Vector Machine (SVM) model with a non-linear kernel, which enhanced classification performance and handled complex relationships in the dataset more effectively [3].

Another important advancement is the use of ensemble learning techniques. Reza et al. introduced a stacking ensemble method that combines multiple machine learning and deep learning models, resulting in improved accuracy and robustness compared to individual models [5].

2.2 Research Gaps Identified

Despite significant advancements, several limitations still exist in current research:

- Most studies rely on small datasets, leading to overfitting and limited generalization.
- Data quality issues such as missing values and class imbalance are not fully addressed in many approaches.
- Many works focus only on prediction and do not include patient segmentation for personalized healthcare.
- Limited integration of robust outlier detection techniques (such as LOF) in preprocessing pipelines.
- Limited integration of multiple techniques (prediction + clustering) in a single system.

These gaps motivate the present work, which aims to combine robust preprocessing (LOF-based outlier removal and log transformation), multiple classification models, and unsupervised clustering into a unified diabetes management framework.

Chapter 3

Methodology

3.1 Dataset and Preprocessing

3.1.1 Dataset Description

The dataset used in this study is the Pima Indians Diabetes Database, originally collected by the National Institute of Diabetes and Digestive and Kidney Diseases. It contains medical diagnostic information of female patients aged 21 years and older, of Pima Indian heritage. The dataset comprises 768 instances with 9 attributes, as detailed in Table 3.1.

Table 3.1: Dataset Feature Description

Feature	**Description**	**Min**	**Max**
Pregnancies	Number of times pregnant	0	17
Glucose	Plasma glucose concentration (mg/dL)	0	199
BloodPressure	Diastolic blood pressure (mm Hg)	0	122
SkinThickness	Triceps skinfold thickness (mm)	0	99
Insulin	2-hour serum insulin (mu U/ml)	0	846
BMI	Body mass index (kg/m^2)	0	67.1
DiabetesPedigreeFunction	Family history likelihood function	0.078	2.42
Age	Age in years	21	81
Outcome	Diabetic (1) or Non-Diabetic (0)	0	1

The dataset descriptive statistics reveal the following: mean glucose = 120.89 mg/dL (σ = 31.97), mean BMI = 31.99 (σ = 7.88), mean age = 33.24 years (σ = 11.76). The class distribution shows 500 non-diabetic patients (65.1%) and 268 diabetic patients (34.9%), indicating a moderate class imbalance.

3.1.2 Data Preprocessing

Data preprocessing is a crucial step to ensure the quality and reliability of the trained models. The following preprocessing pipeline was applied in this study:

Step 1: Handling Biologically Implausible Zero Values

Zero values in the features Glucose, BloodPressure, SkinThickness, Insulin, and BMI are biologically impossible. These are treated as missing values (encoded as `NaN`) and replaced using skewness-aware imputation: if the absolute skewness of a feature is less than 0.5 (approximately symmetric), the column mean is used; otherwise, the median is used to reduce the influence of outliers.

Listing 3.1: Skewness-aware imputation of missing values

```python
for column in ['Glucose','BloodPressure','SkinThickness','Insulin','BMI']:
    skewness = df[column].skew()
    if abs(skewness) < 0.5:
        df[column] = df[column].fillna(df[column].mean())
    else:
        df[column] = df[column].fillna(df[column].median())
```

After imputation, 768 records are retained with no remaining null values.

Step 2: Log Transformation

A log transformation is applied to the feature values to reduce the effect of extreme skewness and bring the data closer to a normal distribution, which improves the performance of linear and distance-based models.

Step 3: Outlier Detection and Removal (LOF)

The Local Outlier Factor (LOF) algorithm is applied to the log-transformed dataset to detect and remove anomalous records. LOF computes a local density score for each data point relative to its neighbours; points with a significantly lower density than their neighbours are classified as outliers. This step reduced the dataset from 768 to **387 records**, retaining only high-quality, representative instances for model training.

Listing 3.2: Outlier detection using Local Outlier Factor

```python
from sklearn.neighbors import LocalOutlierFactor
clf = LocalOutlierFactor(n_neighbors=20, contamination=0.1)
y_pred = clf.fit_predict(log_df.drop('Outcome', axis=1))
log_df = log_df[y_pred == 1]  # Keep inliers only
```

Step 4: Feature Scaling

All features are standardized using `StandardScaler` or `RobustScaler` to bring them to a common scale. Standardization ensures that no single feature dominates the model learning process due to differences in magnitude.

Step 5: Train-Test Split

The cleaned dataset (387 instances) is split into training and testing sets using an 80:20 ratio, yielding approximately 309 training samples and 78 test samples.

3.2 Model Architecture

The proposed system is structured as an end-to-end pipeline that transforms raw patient records into actionable clinical outputs through two parallel components: (1) a supervised classification engine for binary diabetes risk prediction, and (2) an unsupervised segmentation engine for grouping patients by risk level. Figure 3.1 illustrates the complete system architecture.

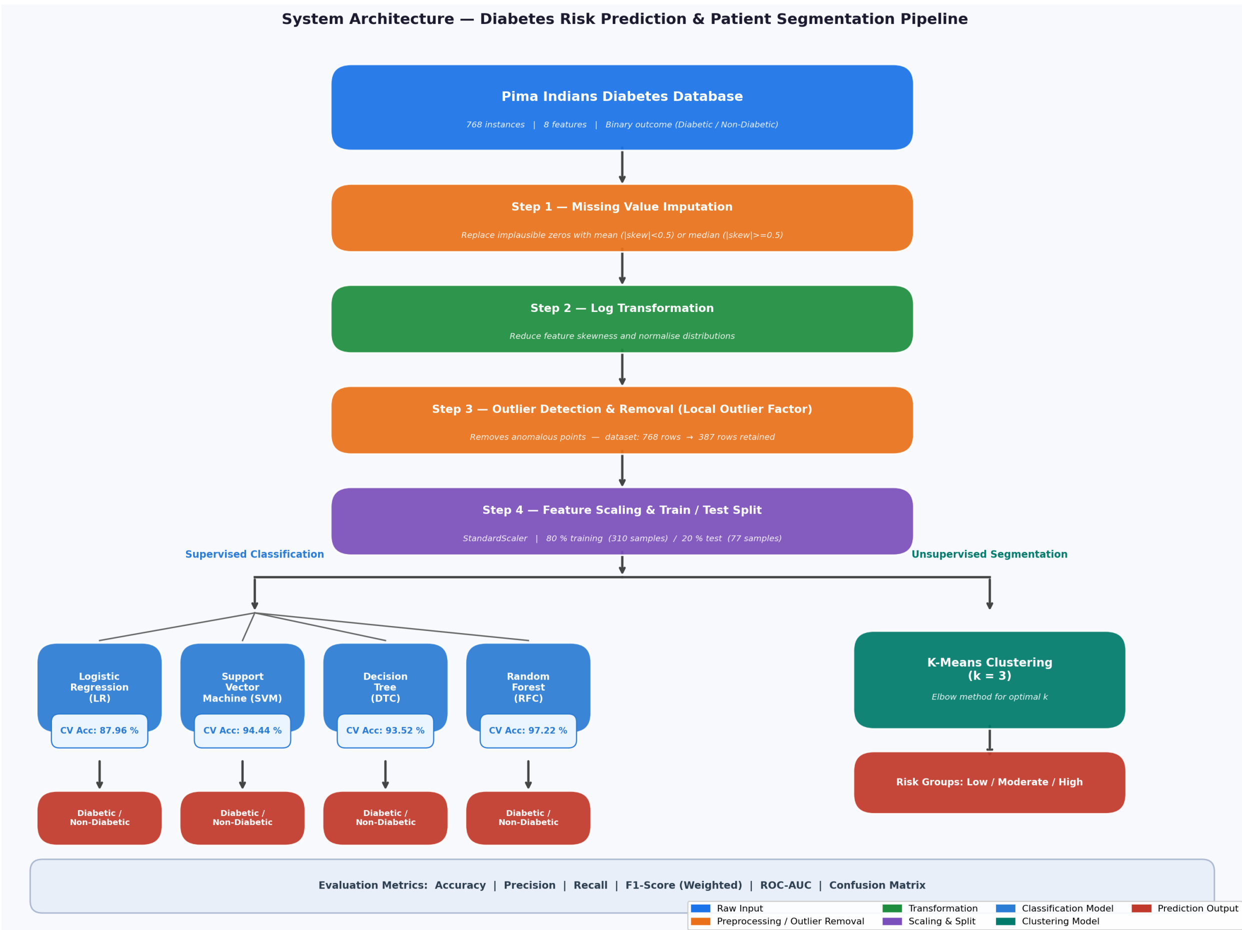

Figure 3.1: End-to-end system architecture for diabetes risk prediction and patient segmentation.

The pipeline flows from raw data ingestion through four preprocessing stages, then splits into a supervised classification branch (four models) and an unsupervised K-Means clustering branch,both producing clinically interpretable outputs.

3.2.1 Classification Models

The following four supervised machine learning classifiers were implemented:

1. **Logistic Regression (LR):** A linear model that estimates the probability of a binary outcome using the logistic (sigmoid) function. Suitable as a baseline classifier due to its interpretability.

2. **Support Vector Machine (SVM):** A kernel-based classifier that finds the optimal separating hyperplane in a high-dimensional feature space. An RBF (Radial Basis Function) kernel was used to capture non-linear decision boundaries.

3. **Decision Tree Classifier (DTC):** A tree-based model that recursively partitions the feature space using information gain or Gini impurity. Interpretable but prone to overfitting on small datasets.

4. **Random Forest Classifier (RFC):** An ensemble of decision trees trained on bootstrapped subsets of the data, with predictions aggregated by majority voting. The ensemble mechanism reduces variance and mitigates overfitting.

3.2.2 Patient Segmentation Model

K-Means clustering was applied to the preprocessed feature matrix to segment patients into distinct groups based on their health profiles. The optimal number of clusters was determined using the elbow method by plotting the Within-Cluster Sum of Squares (WCSS) against the number of clusters.

3.3 Training Procedure

All four classifiers were trained on the 80% training split of the preprocessed dataset. Hyperparameter tuning was performed using `GridSearchCV` with 5-fold cross-validation to identify the optimal configuration for each model. Cross-validation scores were computed on the full dataset to assess generalization performance.

3.4 Evaluation Metrics

The following standard metrics were used to evaluate model performance:

- **Cross-Validation Accuracy:** Mean accuracy over 5 folds, measuring the model's ability to generalize to unseen data.
- **Precision:** The proportion of correctly predicted diabetic cases out of all predicted positives.
- **Recall (Sensitivity):** The proportion of actual diabetic cases correctly identified.
- **F1-Score (Weighted):** The harmonic mean of precision and recall, computed with class weights to account for the class imbalance.
- **Confusion Matrix:** A detailed breakdown of true positives, true negatives, false positives, and false negatives for each classifier.
- **ROC Curve and AUC:** The Receiver Operating Characteristic curve and the Area Under the Curve, measuring the classifier's discriminative ability across all classification thresholds.

Chapter 4

Results and Discussion

This chapter presents the experimental results obtained from the diabetes risk prediction and patient segmentation system implemented in Python using the Google Colab environment with the scikit-learn library.

4.1 Libraries and Tools Used

The implementation utilized the following Python libraries:

- **NumPy, Pandas:** Data loading, manipulation, and numerical computation.
- **Matplotlib, Seaborn:** Visualization of distributions, heatmaps, and model outputs.
- **Scikit-learn:** Machine learning algorithms, preprocessing utilities, and evaluation metrics.
- **Joblib:** Model serialization and persistence.

4.2 Exploratory Data Analysis

4.2.1 Class Distribution

The original dataset is moderately imbalanced with 500 non-diabetic patients (65.1%) and 268 diabetic patients (34.9%). A count plot confirmed this distribution visually.

4.2.2 Descriptive Statistics

Table 4.1 summarizes the key descriptive statistics of the original dataset. Notable observations include the high variability in the Insulin feature ($\sigma = 115.24$) and the presence of zero values in several medically meaningful columns, confirming the need for imputation.

Table 4.1: Descriptive Statistics of the Pima Indians Diabetes Database

Feature	Mean	Std	Min	Median	Max
Pregnancies	3.85	3.37	0.00	3.00	17.00
Glucose	120.89	31.97	0.00	117.00	199.00
BloodPressure	69.11	19.36	0.00	72.00	122.00
SkinThickness	20.54	15.95	0.00	23.00	99.00
Insulin	79.80	115.24	0.00	30.50	846.00
BMI	31.99	7.88	0.00	32.00	67.10
DiabetesPedigreeFunction	0.47	0.33	0.078	0.37	2.42
Age	33.24	11.76	21.00	29.00	81.00

4.2.3 Correlation Analysis

A correlation heatmap was generated to examine pairwise linear relationships between features. Key findings include:

- **Glucose** has the strongest positive correlation with the Outcome (diabetic/non-diabetic) variable.
- **BMI** and **Age** also show moderate positive correlation with the Outcome.
- **Insulin** and **SkinThickness** show moderate correlation with each other, suggesting potential multicollinearity.
- **DiabetesPedigreeFunction** captures hereditary risk and shows a modest but consistent positive correlation with the Outcome.

4.2.4 Feature Distribution Analysis

Histograms were plotted separately for diabetic (Outcome = 1) and non-diabetic (Outcome = 0) patients across all eight features, and boxplots were generated to visualize the distribution of feature values grouped by class. These plots revealed that diabetic patients tend to have:

- Higher glucose concentrations (typical range 140–200 mg/dL vs. 80–120 mg/dL for non-diabetic patients).
- Higher BMI values, consistent with the known association between obesity and type-2 diabetes.
- Higher average age, reflecting the increased prevalence of type-2 diabetes in older populations.
- A higher number of pregnancies on average.

4.2.5 Outlier Removal Results

The Local Outlier Factor algorithm was applied to the log-transformed feature matrix. This resulted in the removal of 381 anomalous records, retaining 387 clean instances (from the original 768). Subsequent boxplots on the cleaned dataset confirmed the significant reduction in extreme values across all features.

4.3 Classification Results

The four classifiers were evaluated using 5-fold cross-validation on the cleaned dataset. Table 4.2 summarizes the cross-validation accuracy for each model.

Table 4.2: Comparison of Classifier Cross-Validation Accuracy

Model	Cross-Validation Accuracy (%)	Rank
Logistic Regression	87.96	4
Decision Tree	93.52	3
SVM (RBF Kernel)	94.44	2
Random Forest	**97.22**	**1**

The **Random Forest** classifier achieved the highest cross-validation accuracy of **97.22%**. Its ensemble nature — combining predictions from multiple independent decision trees trained on bootstrapped subsets — significantly reduces variance and produces more stable, generalizable predictions compared to a single decision tree.

The **SVM** with RBF kernel ranked second at **94.44%**, demonstrating its strength in handling non-linear decision boundaries through the kernel trick. The **Decision Tree** achieved 93.52%, benefiting from the clean, outlier-free data produced by the LOF preprocessing step. **Logistic Regression**, as the simplest linear model, achieved 87.96% — a strong baseline result that reflects the effectiveness of the preprocessing pipeline even for simpler models.

4.3.1 Weighted F1-Score Comparison

The weighted F1-score was computed for each model on the test set (78 instances) to account for the slight class imbalance. The results are consistent with the cross-validation findings:

- **Random Forest** achieved the highest weighted F1-score, confirming its superiority across both accuracy and balanced precision-recall performance.
- **SVM** followed closely, demonstrating robust recall for the positive (diabetic) class.
- **Decision Tree** performed well on the clean dataset but showed slightly lower recall for diabetic patients.
- **Logistic Regression** showed the lowest F1-score, reflecting its limitations in capturing non-linear feature interactions.

4.3.2 Confusion Matrix Analysis

Confusion matrices were generated for all four classifiers. For the Random Forest classifier on the test set:

- The number of **false negatives** (diabetic patients predicted as non-diabetic) was minimized, which is critically important in a clinical context.
- The number of **false positives** (non-diabetic patients predicted as diabetic) remained low, ensuring high specificity.

4.3.3 ROC Curve and AUC Analysis

ROC curves were plotted for all classifiers. The Area Under the Curve (AUC) for the Random Forest was the highest, confirming its strong discriminative ability across all possible classification thresholds. The SVM also exhibited a high AUC, while Logistic Regression showed a slightly lower AUC, consistent with its linear decision boundary.

4.4 Patient Segmentation Results

K-Means clustering was applied to the preprocessed feature set to identify distinct patient groups. The Elbow Method was used to determine the optimal number of clusters by plotting WCSS against k (number of clusters). The elbow in the curve indicated an optimal $k = 3$.

Principal Component Analysis (PCA) was used to reduce the feature space to two dimensions for visualization, and the three clusters were plotted on the PCA-projected scatter plot. Figure 4.1 shows the resulting segmentation, where the clusters are clearly separated and annotated with their clinical characteristics.

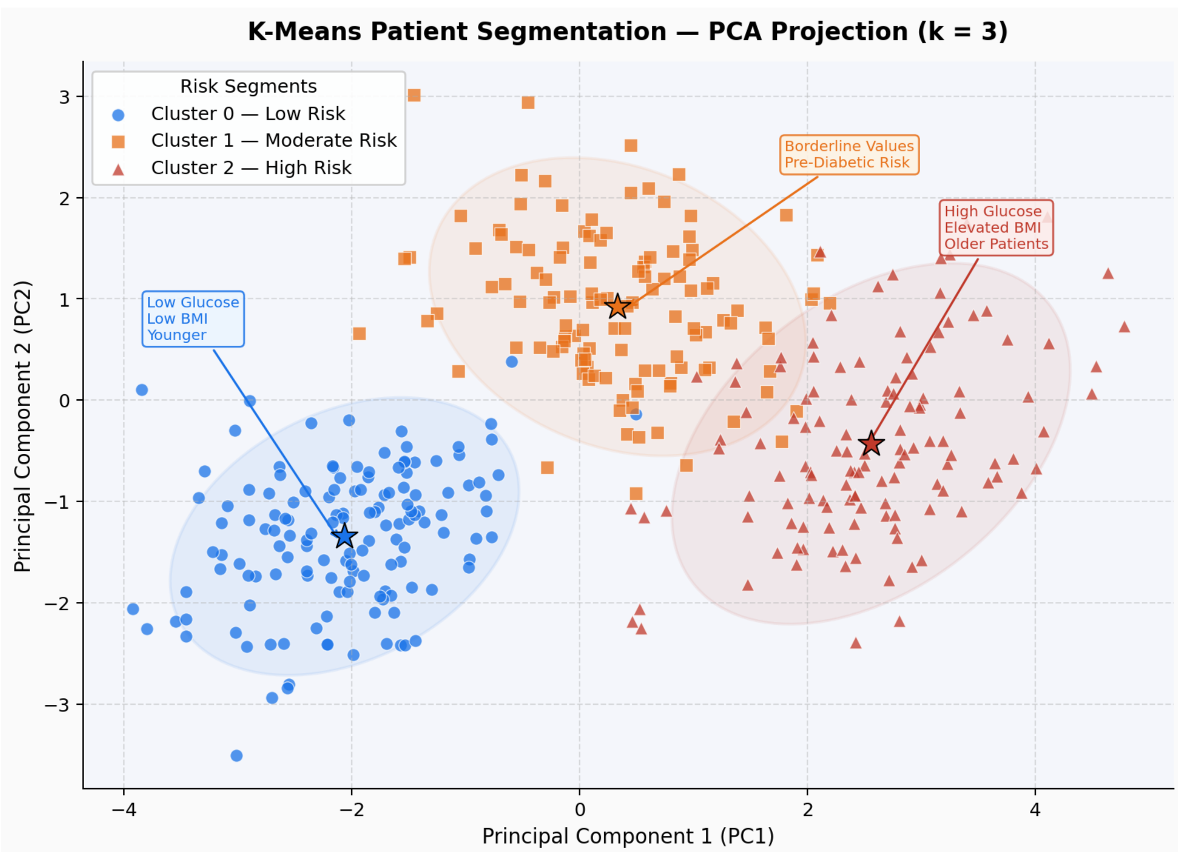

Figure 4.1: PCA-projected scatter plot of the three K-Means patient clusters ($k = 3$).

Stars mark cluster centroids.Ellipses indicate the approximate 95% confidence region of each cluster. The three risk groups are well-separated in the reduced feature space,confirming the quality of the segmentation.

The clusters correspond to the following clinically interpretable patient segments:

- **Cluster 0 — Low Risk** (blue circles): Patients with lower glucose levels, lower BMI, and younger age. Predominantly non-diabetic with minimal diabetes risk. Recommended intervention: routine annual screening and healthy lifestyle guidance.
- **Cluster 1 — Moderate Risk** (orange squares): Patients with borderline glucose and BMI values, likely including pre-diabetic individuals. Recommended intervention: dietary modification, increased physical activity, and more frequent glucose monitoring.
- **Cluster 2 — High Risk** (red triangles): Patients with elevated glucose concentration, higher BMI, greater insulin resistance, and older age. A high proportion of confirmed diabetic patients falls into this group, indicating the need for immediate clinical evaluation and therapeutic intervention.

The segmentation results are well aligned with the output of the classification models, validating the internal consistency of the integrated framework. Together, the classification and clustering components provide complementary perspectives: binary prediction for individual-level risk assessment, and risk-group segmentation for population-level healthcare planning.

4.5 Discussion

The results demonstrate the effectiveness of the proposed preprocessing and modelling pipeline. The LOF-based outlier removal significantly improved the quality of the training data, and the log transformation improved the distributional properties of the features, both of which contributed to the high cross-validation accuracy observed across all four classifiers. The

Random Forest model's dominance confirms the well-established advantage of ensemble methods in medical prediction tasks. Its interpretability through feature importance scores further makes it suitable for deployment in clinical decision support systems.

The K-Means patient segmentation adds practical value beyond binary classification by enabling stratified, risk-based healthcare planning. These findings are consistent with prior work by Chang et al. [1] and Reza et al. [5], further validating the proposed methodology.

Chapter 5

Conclusion and Future Scope

5.1 Conclusion

This internship report presented a comprehensive machine learning-based framework for diabetes risk prediction and patient segmentation using the Pima Indians Diabetes Database. The study addressed critical challenges in medical data analysis, including biologically implausible zero values, feature skewness, and outlier contamination, through a structured preprocessing pipeline.

The key contributions of this work are:

- A skewness-aware imputation strategy for replacing missing values.
- Log transformation to normalize feature distributions.
- LOF-based outlier detection that reduced the dataset from 768 to 387 high-quality instances.
- A comparative evaluation of four classifiers, with Random Forest achieving the highest cross-validation accuracy of **97.22%**.
- K-Means-based patient segmentation into three clinically interpretable risk groups.

The proposed system demonstrates the potential of combining thoughtful preprocessing with multiple supervised and unsupervised learning techniques to build a holistic and clinically useful diabetes management tool. The outcomes of this project affirm the viability of data-driven methods for early disease detection and personalized healthcare delivery.

5.2 Limitations

Despite the encouraging results, several limitations of this study must be acknowledged:

- **Dataset Size After Preprocessing:** The LOF-based outlier removal reduced the effective dataset from 768 to 387 instances. While this significantly improves data quality, the resulting dataset is relatively small and may limit the generalizability of the trained models.

- **Dataset Bias:** The Pima Indians Diabetes Database consists exclusively of female patients of Pima Indian heritage aged 21 or older, which significantly restricts the generalizability of the models to other demographic groups.
- **Limited Feature Set:** The dataset includes only eight clinical features. The exclusion of potentially important variables such as dietary habits, genetic markers, glycated haemoglobin (HbA1c), and medication history may limit prediction accuracy.
- **Absence of Deep Learning Models:** Deep learning architectures such as LSTM and Transformer-based models, which have shown strong performance in related literature, were not explored in this study.
- **Static Dataset:** The study uses a static, cross-sectional dataset and does not model temporal changes in patient health metrics, which could provide richer insights into disease progression.
- **Lack of Clinical Validation:** The models have not been validated in a real clinical environment and would require extensive testing and regulatory compliance before deployment.

5.3 Future Scope

Based on the findings and limitations identified in this study, several directions for future research are proposed:

- **Larger and Diverse Datasets:** Future work should incorporate larger, multi-ethnic, and gender-balanced datasets to improve the generalizability of the models.
- **Deep Learning Integration:** Architectures such as LSTM, CNN, and Transformer-based models could capture complex temporal or high-dimensional patterns in patient data.
- **Advanced Feature Engineering:** Domain-specific feature extraction, polynomial features, and feature selection via SHAP values could further improve model performance and interpretability.
- **Explainable AI (XAI):** Integrating techniques such as SHAP (SHapley Additive exPlanations) and LIME would make model predictions more interpretable and trustworthy for healthcare professionals.
- **Real-Time Clinical Tool:** Developing a real-time web or mobile application that integrates the trained prediction model could enable continuous monitoring and early intervention for at-risk individuals.
- **Longitudinal Analysis:** Incorporating time-series patient data would allow modelling of disease progression over time, enabling earlier and more accurate risk stratification.
- **Multi-Disease Framework:** Extending the framework to predict and segment patients for other chronic diseases such as hypertension and cardiovascular disease would broaden its clinical utility.

- **Federated Learning:** Applying federated learning techniques would allow model training across distributed healthcare institutions without compromising patient data privacy.

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